TECHNICAL REPORT

LUN-TR-20261030-2352D6

LUNARIS

LUNAR UNCHARTED NAVIGATION, AUTONOMY & RESOURCE INTELLIGENCE SYSTEM

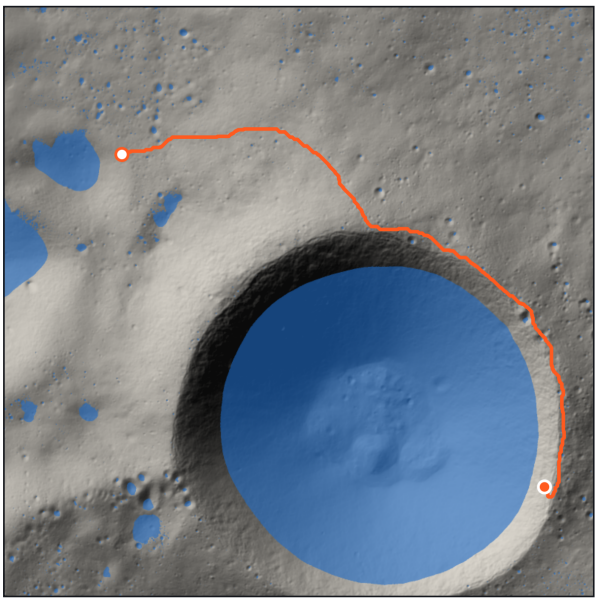
Traverse and Volatile-Prospect Assessment

Shackleton · Lunar South Polar Region

Issued 30 October 2026 · Prepared with the LUNARIS analysis pipeline

DOCUMENT CONTROL

Report number	LUN-TR-20261030-2352D6
Date of issue	2026-10-30 09:15 UTC
Study region	Shackleton
Dataset	Shackleton crater & south pole — LOLA 20 m/px (real data)
Region centre	89.67 S, 129.78 E · D = 20.9 km
Analysis grid	800 x 800 cells @ 40 m
Configuration	0cc97e48aa17f087
Dataset checksum	5de0c23e315fcde4
Primary sources	NASA LRO: LOLA (PGDA), Diviner (PDS), LROC NAC; LOLA illumination models
Distribution	Unrestricted — derived from NASA open data



LOLA shaded relief, PSR (blue) and candidate traverse (orange)

Lunar surface data: NASA/GSFC/ASU/UCLA/MIT

EXECUTIVE SUMMARY

Summary of findings

This report documents an orbital-data assessment of the Shackleton study area using the LUNARIS pipeline. Terrain derived from LOLA laser altimetry classifies 32% of the analysis area as low mobility hazard and 25% as impassable at the adopted slope and roughness thresholds. Permanently shadowed regions occupy 25.7% of the window. A hazard-weighted candidate traverse of 39.4 km was identified, with a maximum slope of 24.8° and no impassable cells crossed. The volatile investigation-priority model flags 23.7% of the area as candidates for further investigation. A traverse simulation of the NASA VIPER (Volatiles Investigating Polar Exploration Rover) returned **MARGINAL**.

ROUTE LENGTH	MAX SLOPE ON ROUTE	PERMANENT SHADOW	ROVER SIMULATION
39.4 km	24.8°	25.7%	MARGINAL

SECTION 01

Introduction and scope

LUNARIS is a decision-support system for permanently shadowed regions (PSRs) near the lunar poles. It combines laser altimetry, thermal-model products, illumination models and orbital imagery into traceable data layers, plans rover traverses over a documented cost surface, and evaluates them with a terramechanics-based rover simulation. Every layer in this report is classified as measured, derived, model estimate or illustrative, and cells without valid data are treated as unknown rather than safe.

Scope: analyses are derived from orbital remote-sensing and model products. Traverses are simulated candidates on a documented cost model; water-ice results are relative investigation priorities, not detections. Results support, and do not replace, mission certification and in-situ confirmation.

Study region. 32 x 32 km window in south polar stereographic coordinates (x -12..20 km, y -16..16 km) containing Shackleton crater, the south pole, and the Shackleton–de Gerlache connecting ridge. Named features are listed only because this window actually covers them. Named features inside the window (IAU gazetteer): Shackleton crater (rim clipped ~0.9 km at the window's lower edge), Lunar south pole, Shackleton–de Gerlache ridge (partial).

SECTION 02

Data sources

Table 1. Input data products and checksums.

Layer	Product	Instrument	Units	SHA-256
elevation	LDEM_80S_20MPP_ADJ	LRO LOLA	m, height relative to 1737.4 km reference sphere	e627615dfe12
elevation uncertainty	LDEM_80S_20MPP_ADJ_ERR	LRO LOLA	m	8d5eae88b9a6
effective resolution	LDEM_80S_20MPP_ADJ_EFFRES	LRO LOLA	m	d629a3694c55
slope reference	LDSM_80S_20MPP_ADJ	LRO LOLA	deg	4947f038f1b8
psr mask	LPSR_80S_20MPP_ADJ	LRO LOLA (illumination modelling on DEM)	1 = PSR, 0 = not PSR	9fcdb10874ae
image	M139729686LE_20MPP	LRO LROC NAC	radiance factor (I/F), as distributed	298e3f0169f0
illumination avg	AVGVISIB_85S_060M_201608	LRO LOLA (illumination modelling)	fraction of time the solar disc is visible (DN x 0.00004)	2776c3ce120c
earth visibility avg	AVGVISIB_85S_060M_201608_EARTH	LRO LOLA (visibility modelling)	fraction of time the Earth disc is visible (DN x 0.00004)	75e7d11d52ed
thermal max	DLRE_PRP_SOUTH.TAB:temp_max	LRO Diviner (DLRE)	K	a11bbc10c16e
thermal avg	DLRE_PRP_SOUTH.TAB:temp_avg	LRO Diviner (DLRE)	K	6c0ad441478e
ice stability depth	DLRE_PRP_SOUTH.TAB:ice_depth	LRO Diviner (DLRE)	m (depth where sublimation < 1 kg m^-2 Gyr^-1; deeper than 2.87 m is encoded as 999)	f76d09b21c1b

Table 2. Layers not ingested for this study region.

Layer	Status
hydrogen	not provided by this dataset

Citation. Barker et al. (2023) PSJ 4:183 doi:10.3847/PSJ/acf3e1; Paige et al. (2010) Science 330:479

SECTION 03

Methods

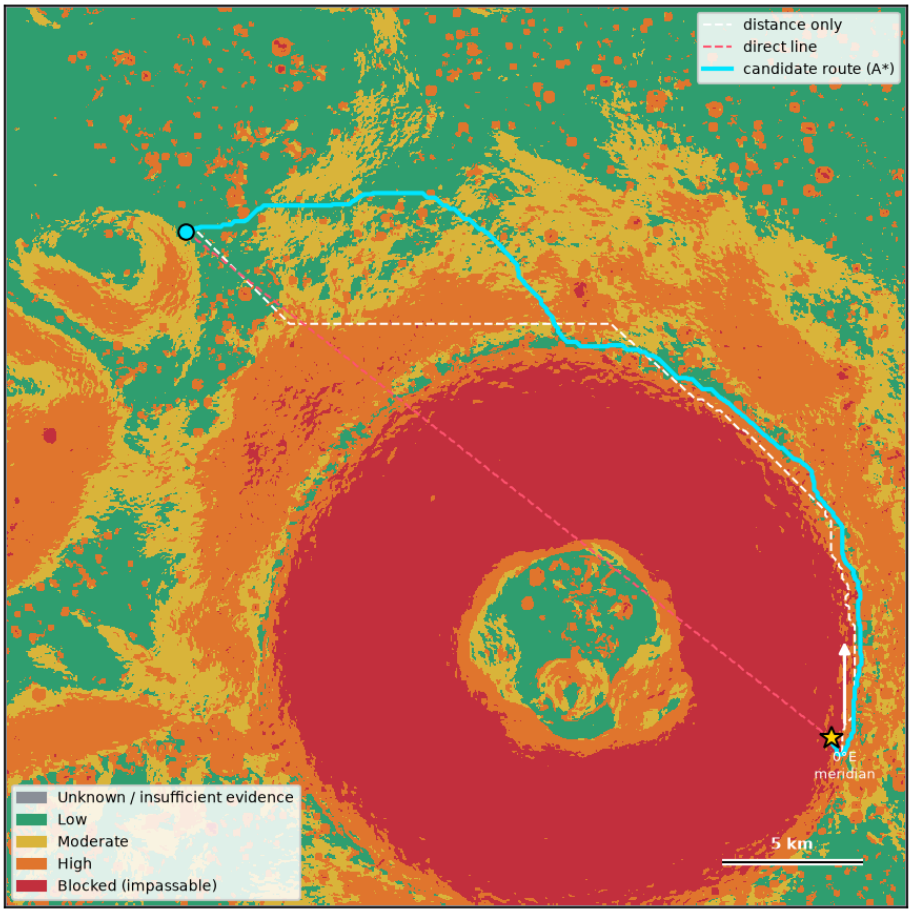
Rasters are ingested with checksum verification, aligned to the elevation grid and block-averaged to the analysis resolution (cells require at least 75% valid sub-cells). Slope uses the Horn (1981) operator; roughness is the RMS residual from a local least-squares plane. Hazard classes take the worse of the slope and roughness classes; producer uncertainty raises a cell by one class. Traverses use an 8-connected A* search with an admissible octile heuristic over a per-metre cost multiplier.

Table 3. Processing record.

Stage	Detail
Ingest	elevation — LDEM_80S_20MPP_ADJ.TIF
Ingest	elevation uncertainty — LDEM_80S_20MPP_ADJ_ERR.TIF
Ingest	effective resolution — LDEM_80S_20MPP_ADJ_EFFRES.TIF
Ingest	slope reference — LDSM_80S_20MPP_ADJ.TIF
Ingest	psr mask — LPSR_80S_20MPP_ADJ.TIF
Ingest	image — M139729686LE_20MPP.TIF
Ingest	illumination avg — sun_avg_visibility.tif
Ingest	earth visibility avg — earth_avg_visibility.tif
Ingest	thermal max — prp_temp_max.tif
Ingest	thermal avg — prp_temp_avg.tif
Ingest	ice stability depth — prp_ice_depth.tif
Align	elevation uncertainty — identical grid (no resampling)
Align	effective resolution — identical grid (no resampling)
Align	slope reference — identical grid (no resampling)
Align	psr mask — identical grid (no resampling)
Align	image — resampled (nearest) onto elevation grid
Align	illumination avg — resampled (bilinear) onto elevation grid
Align	earth visibility avg — resampled (bilinear) onto elevation grid
Align	thermal max — identical grid (no resampling)
Align	thermal avg — identical grid (no resampling)
Align	ice stability depth — identical grid (no resampling)
Resample	factor 2 — nan-aware block mean
Analyse	3.43 s · configuration 0cc97e48aa17f087

SECTION 04

Terrain assessment



Source: Shackleton crater & south pole — LOLA 20 m/px (real data) | Status: derived

Figure 1. Mobility hazard classes with the candidate traverse (cyan), distance-only alternative and direct line. Up is the 0°E meridian direction of the polar stereographic grid.

Table 4. Hazard class distribution over the analysis area.

Hazard class	Area fraction
Unknown / insufficient evidence	1.0%
Low	31.8%
Moderate	16.6%
High	25.3%
Blocked (impassable)	25.3%

Derived slope agrees with the producer slope map with r = 0.998 and a median absolute difference of 0.36° (a consistency check on the same source elevation model).

Boulder-scale hazards. grid is 40 m/px; rover-scale boulders (<10 m) are sub-pixel. Requires native-resolution NAC/ShadowCam imagery (~0.5–2 m/px) and a labelled training/evaluation set.

SECTION 05

Traverse planning

Table 5. Candidate traverse compared with rejected alternatives.

Metric	Candidate (hazard-weighted)	Distance-only	Direct line
Length (m)	39,434	36,984	30,336
Path cost	102,420	193,096	impassable
Maximum slope (°)	24.8	24.9	34.1
Mean slope (°)	7.2	14.4	21.6
Cells in PSR	1	20	330
Blocked cells crossed	0	0	282
Total ascent (m)	2,844	3,141	5,906

Path cost = sum over moves of step length x mean per-metre cost multiplier of the two cells; a value equal to the length means flat, certain, low-hazard terrain all the way. Relative units — not energy, time or risk probability.

Start 89.6789° S, 325.04° E; goal 89.3407° S, 120.08° E (planetocentric).

Table 6. Hazard exposure along the candidate traverse.

Hazard class along route	Cells
Unknown / insufficient evidence	0
Low	535
Moderate	114
High	202
Blocked (impassable)	0

SECTION 06

Volatile investigation priority

The priority model combines psr, thermal_max, ice_stability_depth with explicit weights; layers not used: hydrogen. A cell is assessed only when thermal evidence is present. Assessed fraction: 100.0%.

Table 7. Investigation-priority class distribution.

Priority class	Area fraction
Not assessed (insufficient evidence)	0.0%
Lower relative priority	73.4%
Moderate relative priority	2.9%
Candidate for further investigation	23.7%

Under 9 weight-perturbation scenarios, 31.5% of assessed cells change class at least once; the ranking should be read with this sensitivity in mind.

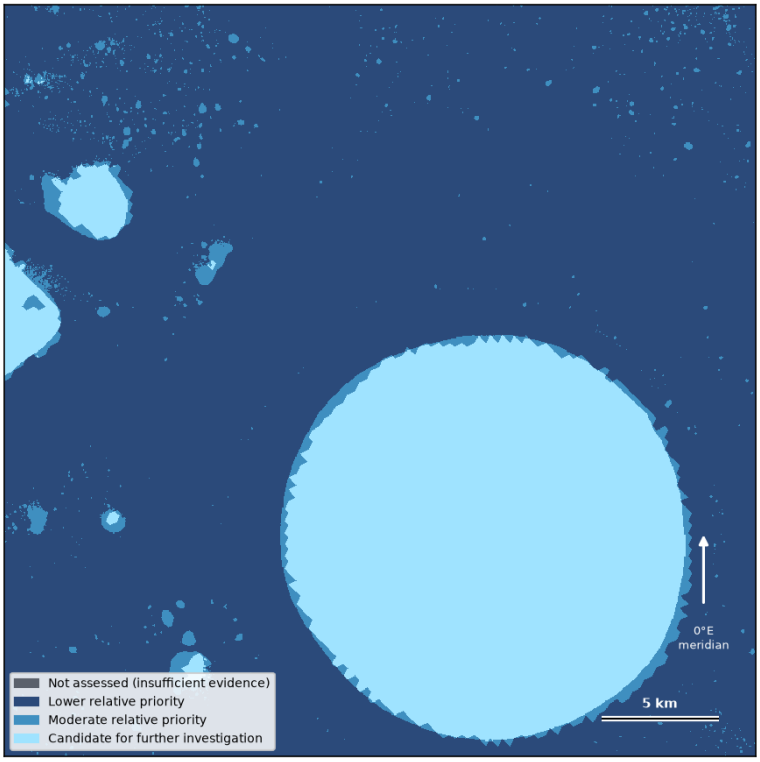


Figure 2. Relative investigation priority for water ice (ranking of evidence, not a detection).

SECTION 07

Rover traverse simulation

The NASA VIPER (Volatiles Investigating Polar Exploration Rover) was driven along the candidate traverse over the native-resolution elevation model on Lunar regolith — Apollo LRV design values, with Bekker–Wong wheel–soil forces, static stability and shadow-time checks. Verdict: **MARGINAL**.

Table 8. Rover simulation results.

Quantity	Value
Distance driven / route length	39.43 / 39.43 km
Drive time (continuous)	128.8 h
Maximum slope / pitch / roll	27.8° / 26.0° / 21.1°
Wheel slip, maximum / mean	74% / 14%
Maximum sinkage	25.3 mm
Minimum static stability margin	7.7°
Time in PSR (longest continuous)	0.23 h (0.17 h)
Shadow energy / usable battery	141 / 4878 Wh
Loose-regolith sensitivity case	FAIL — immobilised: slope 16.3° needs more thrust than the soil can provide at 100 % slip (9.68 km)

Table 9. Simulation events.

Event	Distance (km)
slope 15.9° exceeds the 15° design requirement (1616 m of route)	3.57
slip 43% > 40% operational threshold (308 m of route)	9.68
static stability margin 9.4° < 10°	38.71

Unpublished rover parameters adopted as assumptions: ground clearance m, cg height m, drive efficiency, battery reserve frac. Terrain is the LOLA DEM at its native 20 m grid (bilinear). Rocks, boulders and craters smaller than ~20-40 m are invisible to this simulation; VIPER's published 15 cm obstacle limit cannot be checked from this data. Wheel-soil forces use the Bekker-Wong model with Apollo LRV soil parameters (Costes et al. 1972). Real PSR regolith properties are unmeasured and may differ; see the loose-soil sensitivity run. Quasi-static model: no dynamics, wheel-level suspension articulation, slip-sinkage or turn-in-place time.

SECTION 08

Assumptions

- Slope classes [10.0, 15.0, 25.0] deg and blocked >= 25.0 deg are prototype assumptions.
- Roughness high >= 2.0 m, blocked >= 6.0 m RMS (prototype).
- Cost weights {'slope': 8.0, 'roughness': 3.0, 'uncertainty': 4.0, 'psr': 1.5} are prototype choices; cost is relative, not energy or time.
- Ice-priority weights {'psr': 0.35, 'thermal_max': 0.4, 'ice_stability_depth': 0.25} with sensitivity +/-50%.
- Ground distance equals grid distance (polar stereographic scale error < 0.04 % within 2 deg of the pole).
- No-data and unknown cells are treated as blocked for routing.

SECTION 09

Limitations

- Analysis grid 40.0 m/px: rover-scale obstacles (boulders, small craters) are not resolved.
- PSR map, temperatures and ice-stability depth are model products, each with its own epoch and resolution.
- No LEND hydrogen layer was used; no direct measurement of water ice is included.
- Illumination, communications line-of-sight and thermal survival are not modelled in the route.
- Layer 'LEND hydrogen' unavailable: not provided by this dataset

APPENDIX A

Layer catalogue

Table 10. Analysis layers, epistemic status and method.

Layer	Status	Units	Method
Elevation (LOLA)	measured	m	Block-mean 2x2 of source DEM (cells need >=75% valid sub-pixels).
Hillshade (visual aid)	illustrative	unitless	Lambertian shading of Horn gradients.
Slope	derived	degrees	Horn (1981) finite differences on the analysis grid; NaN propagates from no-data.
Roughness (detrended RMS)	derived	m	Least-squares plane removal per window (separable correlation).
DEM uncertainty (PGDA)	derived	m	Block mean.
DEM effective resolution	derived	m	Block mean.
Elevation flagged uncertain	derived	boolean	Threshold on producer uncertainty layers (prototype thresholds).
Slope — PGDA reference	derived	degrees	Block mean of 20 m slope.
Permanently shadowed regions	derived	fraction of cell	Block mean of the producer's 0/1 PSR map (illumination modelling on LOLA topography).
Annual max temperature (Diviner PRP)	derived	K	Thermal-model fit (PDS PRP) resampled nearest-facet then block-averaged.
Annual mean temperature, 2 cm depth (Diviner PRP)	derived	K	Thermal-model fit (PDS PRP) resampled nearest-facet then block-averaged.
Ice stability depth (Diviner PRP)	derived	m	Thermal-model fit (PDS PRP) resampled nearest-facet then block-averaged.
Average sun visibility (18.6-yr)	derived	fraction of time	LOLA horizon modelling, hourly over 18.6 yr; 60 m product resampled bilinearly, then block mean.
Average Earth visibility (18.6-yr)	derived	fraction of time	LOLA horizon modelling, hourly over 18.6 yr; 60 m product resampled bilinearly, then block mean.
LEND hydrogen	unavailable	-	-
Mobility hazard class	derived	class	Slope edges [10.0, 15.0, 25.0] deg, blocked >= 25.0 deg; roughness high >= 2.0 m, blocked >= 6.0 m.
Traversal cost multiplier	derived	cost per metre (>= 1)	See core/navigation/cost.py
Ice investigation priority (score)	model_estimate	0–1 relative	core/ice_priority/evidence.py
Ice investigation priority (class)	model_estimate	class	Thresholds at 1/3 and 2/3.
Evidence coverage	derived	fraction of weight	-
Priority sensitivity (std)	derived	score std	One-at-a-time perturbation.
Image (LROC NAC)	measured	radiance factor (I/F), as distributed	Block mean for display.

Prepared with LUNARIS (Lunar Uncharted Navigation, Autonomy & Resource Intelligence System). Configuration 0cc97e48aa17f087; dataset checksum 5de0c23e315fcde4; generated 2026-10-30T09:15:55 UTC.